WORKED SOLUTION

STEP III 2010, Question 2

Pure • Definite Integrals

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s3-q2

The key idea

Every denominator here is a quadratic (in x, or in x^2) whose roots turn out to be $\pm e^a$ and $\pm e^{-a}$. Writing the hyperbolic coefficient in exponentials makes the factorisation visible at once, because e^a and e^{-a} have product 1 and sum $2\cosh a$ (or difference $2\sinh a$). After that, partial fractions reduce everything to logarithms or arctangents, and the work is in tidying the answer into hyperbolic form.

Solution

Part (i)

By definition,

$$\cosh a = \tfrac{1}{2}\left(e^a + e^{-a}\right).$$

So $2\cosh a = e^a + e^{-a}$ and $e^a \cdot e^{-a} = 1$, which means

$$x^2 + 2x\cosh a + 1 = (x + e^a)(x + e^{-a}).$$

Both factors are positive for $x \geq 0$, so the integrand is continuous on $[0, 1]$. Since $a > 0$, the two roots $-e^a$ and $-e^{-a}$ are distinct, and the partial fractions are

$$\frac{1}{(x+e^{-a})(x+e^a)} = \frac{1}{e^a - e^{-a}}\left(\frac{1}{x+e^{-a}} - \frac{1}{x+e^a}\right) = \frac{1}{2\sinh a}\left(\frac{1}{x+e^{-a}} - \frac{1}{x+e^a}\right),$$

as you can check by recombining: the numerator becomes $(x+e^a) - (x+e^{-a}) = e^a - e^{-a}$. Integrating,

$$\begin{aligned}\int_0^1 \frac{\mathrm{d}x}{x^2 + 2x\cosh a + 1} &= \frac{1}{2\sinh a}\left[\ln\frac{x+e^{-a}}{x+e^a}\right]_0^1 \\ &= \frac{1}{2\sinh a}\left(\ln\frac{1+e^{-a}}{1+e^a} - \ln\frac{e^{-a}}{e^a}\right).\end{aligned}$$

The first logarithm simplifies nicely: $1+e^{a}=e^{a}(e^{-a}+1)$, so $\dfrac{1+e^{-a}}{1+e^{a}}=e^{-a}$. The second is $\ln e^{-2a}=-2a$. Hence

$$\int_0^1 \frac{\mathrm{d}x}{x^2+2x\cosh a+1}=\frac{1}{2\sinh a}\left(-a+2a\right)=\frac{a}{2\sinh a},$$

which is the required result.

Part (ii)

The first integral. Now the coefficient is $2\sinh a=e^{a}-e^{-a}$ and the constant term is $-1=e^{a}\cdot(-e^{-a})$, so

$$x^2+2x\sinh a-1=(x+e^{a})(x-e^{-a}).$$

Ask yourself Is the integrand well behaved on the whole range $[1,\infty)$?

The roots are $-e^{a}<0$ and e^{-a}, and $e^{-a}<1$ because $a>0$. So for $x\geq 1$ both factors are positive and the integrand is continuous. For large x it behaves like $1/x^2$, so the improper integral converges. Partial fractions as before, now with $e^{a}-(-e^{-a})=2\cosh a$:

$$\frac{1}{(x-e^{-a})(x+e^{a})}=\frac{1}{2\cosh a}\left(\frac{1}{x-e^{-a}}-\frac{1}{x+e^{a}}\right).$$

Integrating from 1 to X and letting $X\to\infty$,

$$\int_1^X \frac{\mathrm{d}x}{x^2+2x\sinh a-1}=\frac{1}{2\cosh a}\left(\ln\frac{X-e^{-a}}{X+e^{a}}-\ln\frac{1-e^{-a}}{1+e^{a}}\right).$$

As $X\to\infty$ the fraction $\frac{X-e^{-a}}{X+e^{a}}\to 1$, so the first logarithm tends to 0. For the second, take out e^{-a} from the numerator:

$$\frac{1-e^{-a}}{1+e^{a}}=e^{-a}\,\frac{e^{a}-1}{e^{a}+1}=e^{-a}\tanh\tfrac{a}{2},$$

using $\dfrac{e^{a}-1}{e^{a}+1}=\dfrac{e^{a/2}-e^{-a/2}}{e^{a/2}+e^{-a/2}}=\tanh\frac{a}{2}$ (divide top and bottom by $e^{a/2}$). So the second logarithm is $-a+\ln\tanh\frac{a}{2}$, and

$$\int_1^\infty \frac{\mathrm{d}x}{x^2+2x\sinh a-1}=\frac{a-\ln\tanh\frac{a}{2}}{2\cosh a}=\frac{a+\ln\coth\frac{a}{2}}{2\cosh a}.$$

Both terms in the numerator are positive (since $0<\tanh\frac{a}{2}<1$), as they must be for a positive integrand.

The second integral. This is the quadratic of part (i) with x replaced by x^2:

$$x^4+2x^2\cosh a+1=(x^2+e^{a})(x^2+e^{-a}),$$

and both factors are positive for all real x, so the integrand is continuous, and it is $O(1/x^4)$ at infinity, so the integral converges. Treating x^2 as the variable in the same partial fractions,

$$\frac{1}{(x^2+e^{-a})(x^2+e^{a})} = \frac{1}{2\sinh a}\left(\frac{1}{x^2+e^{-a}} - \frac{1}{x^2+e^{a}}\right).$$

Each piece is a standard arctangent integral: for $c>0$,

$$\int_0^\infty \frac{\mathrm{d}x}{x^2+c^2} = \left[\frac{1}{c}\arctan\frac{x}{c}\right]_0^\infty = \frac{\pi}{2c}.$$

With $c = e^{-a/2}$ and $c = e^{a/2}$ respectively (each piece converges on its own, so we may split the integral),

$$\int_0^\infty \frac{\mathrm{d}x}{x^4+2x^2\cosh a+1} = \frac{1}{2\sinh a}\cdot\frac{\pi}{2}\left(e^{a/2}-e^{-a/2}\right) = \frac{\pi}{2}\cdot\frac{2\sinh\frac{a}{2}}{2\sinh a}.$$

Finally, the double-angle formula $\sinh a = 2\sinh\frac{a}{2}\cosh\frac{a}{2}$ cancels the $\sinh\frac{a}{2}$:

$$\int_0^\infty \frac{\mathrm{d}x}{x^4+2x^2\cosh a+1} = \frac{\pi}{4\cosh\frac{a}{2}}.$$

Ask yourself Is there a way to see the $\cosh\frac{a}{2}$ coming?

Yes. Call the integral I. The substitution $x \mapsto 1/x$ shows $I = \int_0^\infty \frac{x^2}{x^4+2x^2\cosh a+1}\,\mathrm{d}x$ as well, so

$$2I = \int_0^\infty \frac{1+1/x^2}{x^2+1/x^2+2\cosh a}\,\mathrm{d}x.$$

With $u = x - 1/x$ we have $\mathrm{d}u = (1+1/x^2)\,\mathrm{d}x$ and $x^2+1/x^2 = u^2+2$, while u runs from $-\infty$ to ∞. The denominator becomes $u^2+2+2\cosh a = u^2+4\cosh^2\frac{a}{2}$, so $2I = \pi/(2\cosh\frac{a}{2})$, the same answer.

Checks

- As $a\to 0$, part (i) becomes $\int_0^1 (1+x)^{-2}\,\mathrm{d}x = \frac{1}{2}$, and $a/(2\sinh a)\to\frac{1}{2}$. The last integral becomes $\int_0^\infty (1+x^2)^{-2}\,\mathrm{d}x = \frac{\pi}{4}$, matching $\pi/(4\cosh 0)$.
- Numerically, with $a=1$, Simpson's rule gives 0.4254590641 for part (i), 0.5741556186 for the first integral of (ii) and 0.6965059227 for the second, agreeing with $\frac{1}{2\sinh 1}$, $\frac{1-\ln\tanh\frac{1}{2}}{2\cosh 1}$ and $\frac{\pi}{4\cosh\frac{1}{2}}$ to ten decimal places.

Common slips

- Getting the partial-fraction constant wrong: it is $1/(\text{difference of the roots})$, which is $1/(2\sinh a)$ in part (i) but $1/(2\cosh a)$ in the first integral of (ii).

- Forgetting to check that $e^{-a} < 1$, so that the root of the denominator lies outside $[1, \infty)$.
- Leaving the answers in exponentials: the question asks for hyperbolic functions, so $\frac{e^a-1}{e^a+1}$ should become $\tanh \frac{a}{2}$.

Beyond STEP

The second integral of (ii) is usually done at university by contour integration: the integrand has simple poles at $\pm ie^{\pm a/2}$, and summing residues in the upper half-plane gives $\pi/(4\cosh \frac{a}{2})$ in a few lines. The same method handles $\int_0^\infty \mathrm{d}x/(1+x^n)$ and many integrals that resist real methods. Needham's *Visual Complex Analysis* gives a geometric account of residues and why they work.

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